Arujuna JEE 2024

Motion in a straight line (PYQ)

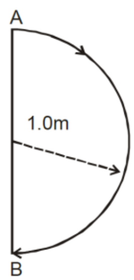
1. A particle is moving with speed $v = d\sqrt{x}$ along positive x-axis. Calculate the speed of the particle at time t = τ (assume that the particle is at origin at t = 0). **[Main 12Apr. 2019II]**

(A) $\frac{b^2\tau}{4}$ (B) $\frac{b^2\tau}{2}$

(C) $b^2\tau$ (D) $\frac{b^2\tau}{\sqrt{2}}$

2. In 1.0 s, a particle goes from point A to point B, moving in a semicircle of radius 1.0 m (see Figure). The magnitude of the average velocity A

[1999S - 2 Marks]



(A) 3.14 m/s (B) 2.0 m/s

(C) 1.0 m/s (D) Zero

3. A particle is moving eastwards with a velocity of 5 m/s. In 10 s the velocity changes to 5 m/s northwards. The average acceleration in this time is/are **[1982 - 3 Marks]**

(A) Zero

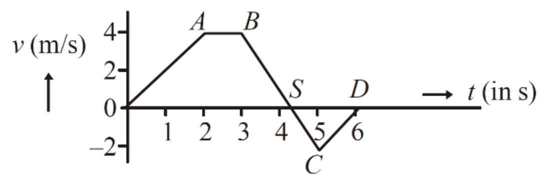
(B) $1/\sqrt{2}$ m/s^2 towards north-west

(C) $1/\sqrt{2}$ m/s^2 towards north-east

(D) 1/2 m/s^2 towards north-west

(E) 1/2 m/s^2 towards north

4. The velocity (v) and time (t) graph of a body in a straight line motion is shown in the figure. The point S is at 4.333 seconds. The total distance covered by the body in 6s is: **[Main 05 Sep. 2020 (II)]**



(A) 37/3 m (B) 12 m

(C) 11 m (D) 49/4 m

5. A bullet of mass 20 g has an initial speed of 1 ms^{-1}, just before it starts penetrating a mud wall of thickness 20cm. If the wall offers a mean resistance of 2.5×10^{-2}N, the speed of the bullet after emerging from the other side of the wall is close to:

[Main 10 Apr. 2019 (II)]

(A) 0.1 ms^{-1} (B) 0.7 ms^{-1}

(C) 0.3 ms^{-1} (D) 0.4 ms^{-1}

6. The position of a particle as a function of time t, is given by $x(t) = at + bt^2 - ct^3$ where, a, b and c are constants. When the particle attains zero acceleration, then its velocity will be:

[Main 9 Apr. 2019 (II)]

(A) $a + \frac{b^2}{4c}$ (B) $a + \frac{b^2}{3c}$

(C) $a + \frac{b^2}{c}$ (D) $a + \frac{b^2}{2c}$

7. The velocity-time graphs of a car and a scooter are shown in the figure. (i) the difference between the distance travelled by the car and the scooter in 15 s and (ii) the time at which the car will catch up with the scooter are, respectively

[Main Online April 15, 2018]

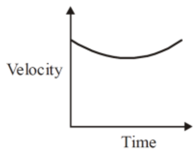
(A) 337.5 m and 25 s

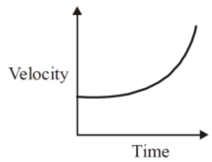
(B) 225.5 m and 10 s

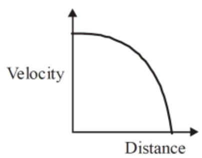
(C) 112.5 m and 22.5 s

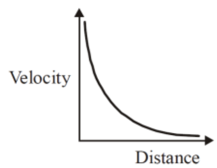
(D) 11.2.5 m and 15 s

8. Which graph corresponds to an object moving with a constant negative acceleration and a positive velocity? **[Main Online April 8, 2017]**

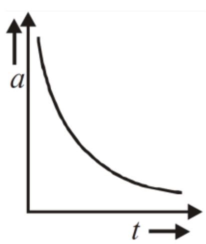
(A) 

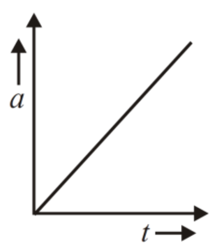
(B) 

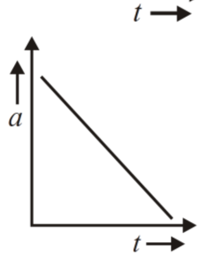
(C) 

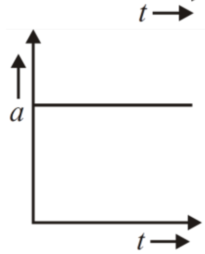
(D) 

9. The distance travelled by a body moving along a line in time t is proportional to t^3. The acceleration-time (a, t) graph for the motion of the body will be **[Main Online May 12, 2012]**

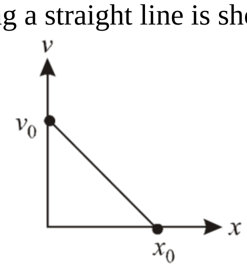
(A) 

(B) 

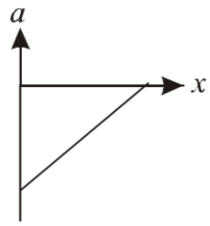
(C) 

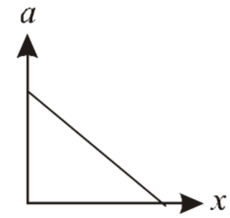
(D) 

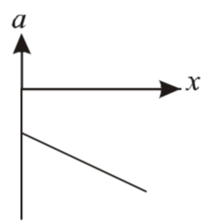
10. The velocity-displacement graph of a particle moving along a straight line is shown **[2005S]**

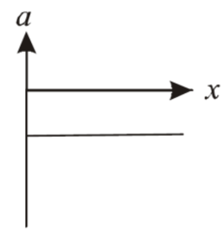


The most suitable acceleration -displacement graph will be

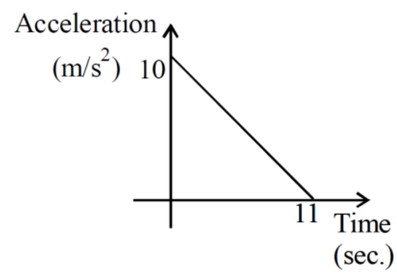
(A) 

(B) 

(C) 

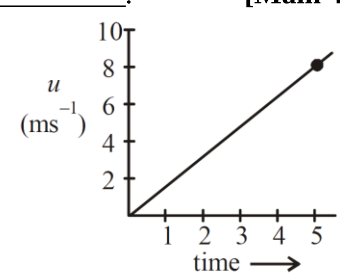
(D) 

11. A body starts from rest at time t = 0, the acceleration time graph is shown in the figure. The maximum velocity attained by the body will be **[2004 S]**



(A) 110 m/s
(B) 55 m/s
(C) 650 m/s
(D) 550 m/s

12. The speed verses time graph for a particle is shown in the figure. The distance travelled (in m) by the particle during the time interval t = 0 to t = 5 s will be ______________. **[Main 4 Sep. 2020 (II)]**



13. The distance x covered by a particle in one dimensional motion varies with time t as $x^2 = at^2 + 2bt + c$. If the acceleration of the particle depends on x as x⁻n, where n is an integer, the value of n is **[Main 9 Jan 2020 (I)]**

14. Train A and train B are running on parallel tracks in the opposite directions with speeds of 36 km/hour and 72 km/hour, respectively. A person is walking in train A in the direction opposite to its motion with a speed of 1.8km/hour. Speed (in ms^{-1}) of this person as observed from train B will be close to : (take the distance between the tracks as negligible)

[Main 2 Sep. 2020(I)]

(A) 29.5 ms^{-1} (B) 28.5 ms^{-1}
(C) 31.5 ms^{-1q} (D) 30.5 ms^{-1}

15. A car is standing 200 m behind a bus, which is also at rest. The two start moving at the same instant but with different forward accelerations. The bus has acceleration 2 m/s^2 and the car has acceleration 4 m/s^2. The car will catch up with the bus after a time of : **[Main Online April 9,2017]**

(A) $\sqrt{110}\,s$ (B) $\sqrt{120}\,s$
(C) $10\sqrt{2}\,s$ (D) 15 s

16. A helicopter rises from rest on the ground vertically upwards with a constant acceleration g. A food packet is dropped from the helicopter when it is at a height h. The time taken by the packet to reach the ground is close to [g is the acceleration due to gravity] : **[Main 5 Sep. 2020 (I)]**

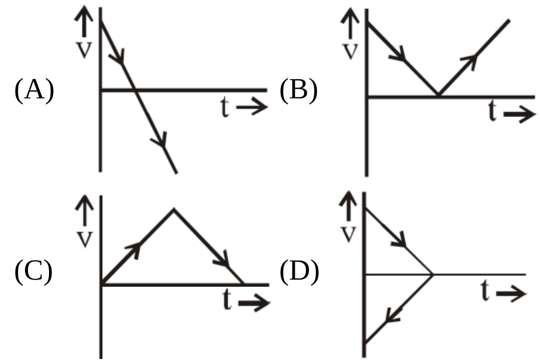
(A) $t=\frac{2}{3}\sqrt{\left(\frac{h}{g}\right)}$

(B) $t=1.8\sqrt{\frac{h}{g}}$

(C) $t=3.4\sqrt{\left(\frac{h}{g}\right)}$

(D) $t=\sqrt{\frac{2h}{3g}}$

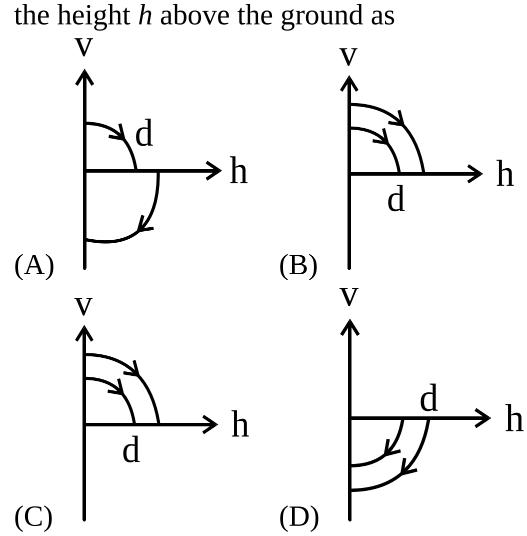
17. A body is thrown vertically upwards. Which one of the following graphs correctly represent the velocity vs time? **[Main 2017]**



18. From a tower of height H, a particle is thrown vertically upwards with a speed u. The time taken by the particle, to hit the ground, is n times that taken by it to reach the highest point of its path. The relation between H, and n is: **[Main 2014]**

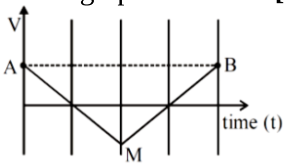
(A) $2gH = n^2u^2$
(B) $gH = (n-2)^2u^2d$
(C) $2gH = nu^2 (n-2)$
(D) $gH = (n-2)u^2$

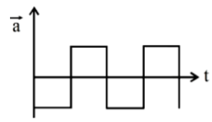
19. A ball is dropped vertically from a height *d* above the ground. It hits the ground and bounces up vertically to a height *d/2*. Neglecting subsequent motion and air resistance, its velocity *v* varies with the height *h* above the ground as **[2000S]**

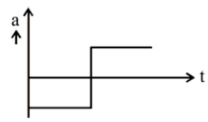


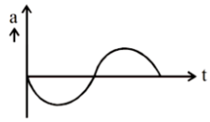
20. A ball is dropped from the top of a 100 m high tower on a planet. In the last $\frac{1}{2}$s before hitting the ground, it covers a distance of 19 m. Acceleration due to gravity (in ms^{-2}) near the surface on that planet is ___________. **[Main 8 Jan. 2020 (II)]**

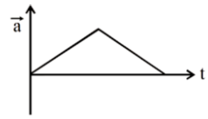
21. If the velocity-time graph has the shape AMB, what would be the shape of the corresponding acceleration-time graph? **[JEE Main 2021]**



(A)


(B)


(C)


(D)


22. A particle is projected with velocity v_0 along x-axis. A damping force is acting on the particle which is proportional to the square of the distance from the origin i.e., $ma = -\alpha x^2$. The distance at which the particle stops : **[JEE Main 2021]**

(A) $\left(\frac{3v_0^2}{2\alpha}\right)^{\frac{1}{2}}$ (B) $\left(\frac{2v_0}{3\alpha}\right)^{\frac{1}{3}}$

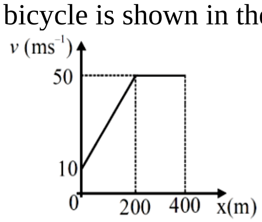
(C) $\left(\frac{2v_0^2}{3\alpha}\right)^{\frac{1}{2}}$ (D) $\left(\frac{3v_0^2}{2\alpha}\right)^{\frac{1}{3}}$

23. An engine of a train, moving with uniform acceleration, passes the signal-post with velocity u and the last compartment with velocity v. The velocity with which middle point of the train passes the signal post is: **[JEE Main 2021]**

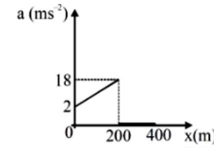
(A) $\sqrt{\frac{v^2+u^2}{2}}$ (B) $\frac{v-u}{2}$

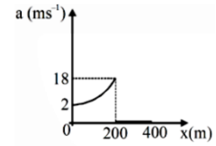
(C) $\frac{u+v}{2}$ (D) $\sqrt{\frac{v^2-u^2}{2}}$

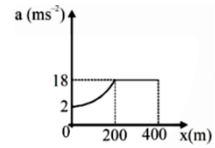
24. The velocity-displacement graph describing the motion of a bicycle is shown in the figure.

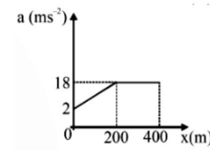


The acceleration-displacement graph of the bicycle's motion is best described by : **[JEE Main 2021]**

(A)


(B)


(C)


(D)


25. A mosquito is moving with a velocity $\vec{v} - = 0.5t^2\hat{i} + 3t\hat{j} + 9\hat{k}\,m/s$ and accelerating in uniform conditions. What will be the direction of mosquito after 2s ? **[JEE Main 2021]**

(A) $\tan^{-1}\left(\frac{2}{3}\right)$ from x-axis

(B) $\tan^{-1}\left(\frac{2}{3}\right)$ from y-axis

(C) $\tan^{-1}\left(\frac{5}{2}\right)$ from y-axis

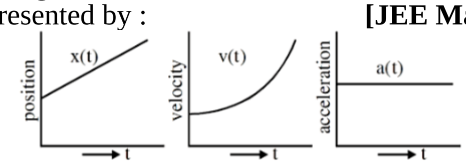
(D) $\tan^{-1}\left(\frac{5}{2}\right)$ from x-axis

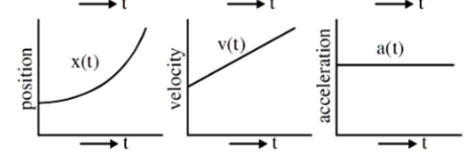
26. The velocity of a particle is $v = v_0 + gt + Ft^2$. Its position is x = 0 at t = 0; then its displacement after time (t 1) is : **[JEE Main 2021]**

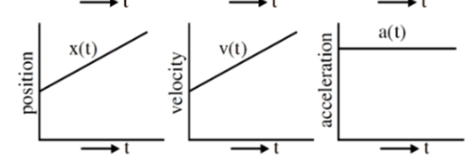
(A) $v_0 + g + F$ (B) $V_0 + \frac{g}{2} + \frac{F}{3}$

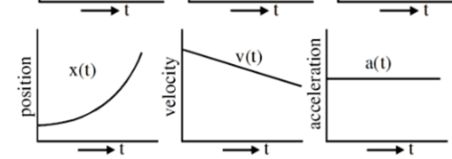
(C) $V_0 + \frac{g}{2} + F$ (D) $v_0 + 2g + 3F$

27. The position, velocity and acceleration of a particle moving with a constant acceleration can be represented by : **[JEE Main 2021]**

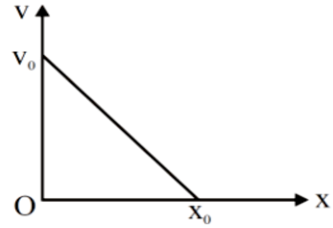
(A)


(B)


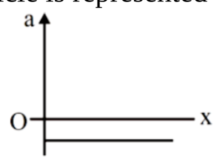
(C)


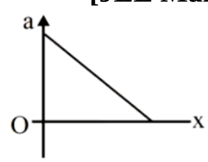
(D)


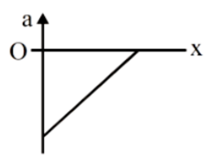
28. The velocity-displacement graph of a particle is shown in the figure.

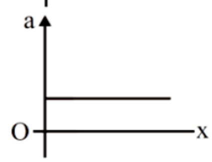


The acceleration-displacement graph of the same particle is represented by : **[JEE Main 2021]**

(A) 

(B) 

(C) 

(D) 

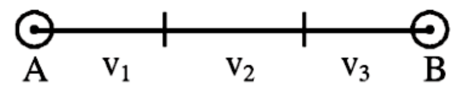
29. The instantaneous velocity of a particle moving in a straight line is given as $v = \alpha t + \beta t^2$, where α and β are constants. The distance travelled by the particle between 1s and 2s is : **[JEE Main 2021]**

(A) $3\alpha + 7\beta$ (B) $\frac{3}{2}\alpha + \frac{7}{3}\beta$

(C) $\frac{\alpha}{2} + \frac{\beta}{3}$ (D) $\frac{3}{2}\alpha + \frac{7}{2}\beta$

30. A ball of mass 0.5 kg is dropped from the height of 10 m. The height, at which the magnitude of velocity becomes equal to the magnitude of acceleration due to gravity, is ……….m. (Use $g = 10\ m/s^2$) **[JEE Main 2022]**

31. A car covers AB distance with first one-third at velocity v_1, ms^{-1}, second one-third at $v_2\ ms^{-1}$ and last one-third at $v_3\ ms^{-1}$. If $v_3 = 3v_1$, $v_2 = 2v_1$ and $v_1 = 11\ ms^{-1}$ then the average velocity of the car is ________ ms^{-1}. **[JEE Main 2022]**



32. A small toy starts moving from the position of rest under a constant acceleration. If it travels a distance of 10m in t s,. the distance travelled by the toy in the next t s will be : **[JEE Main 2022]**

(A) 10 m (B) 20 m

(C) 30 m (D) 40 m

Answer Key

1. (B)
2. (B)
3. (B)
4. (A)
5. (B)
6. (B)
7. (C)
8. (C)
9. (B)
10. (A)
11. (B)
12. (20)
13. (3)
14. (A)
15. (C)
16. (C)
17. (A)
18. (C)
19. (A)
20. (8.00)
21. (B)
22. (D)
23. (A)
24. (A)
25. (B)
26. (B)
27. (B)
28. (C)
29. (B)
30. (5)
31. (18)
32. (C)